

Features of JavaScript

Core Features:

Interpreted Language - Executed line by line

Dynamic Typing - Variables don't need type declaration

Prototype-based Object Orientation - Uses prototypes instead of classes (though ES6 added classes)

First-class Functions - Functions can be assigned to variables, passed as arguments

Event-driven - Supports event listeners and callbacks

Asynchronous Programming - Callbacks, promises, async/await

Detailed Explanation of Two Features

1. First-class Functions

Functions in JavaScript are treated like any other variable. They can be:

Assigned to variables

Passed as arguments to other functions

Returned from functions

Stored in data structures

```
e.g const greet = function(name) {  
  return `Hello, ${name}!`;  
};
```

```
// Function passed as argument  
function executeFunction(fn, value) {  
  return fn(value);  
}
```

```
console.log(executeFunction(greet, "Alice"));
```

2. Asynchronous Programming

JavaScript handles time-consuming operations (like API calls, file reading) without blocking the main thread.

Evolution:

Callbacks (Original method, can lead to "callback hell")

Promises (ES6 - better error handling and chaining)

Async/Await (ES2017 - makes async code look synchronous)

Example with Promises and Async/Await:

```
fetch('https://api.example.com/data')  
.then(response => response.json())  
.then(data => console.log(data))  
.catch(error => console.error('Error:', error));
```

Types of Comments in JavaScript

JavaScript supports two types of comments:

1. Single-line Comments

Syntax: `// comment text`

Everything after `//` on that line is ignored

Can be at the end of a code line or on a separate line

javascript

`// This is a single-line comment at the beginning`

`let age = 25; // This comment is after code`

`// let name = "John"; // This line is commented out entirely`

2. Multi-line Comments (Block Comments)

Syntax: `/ comment text /`

Everything between `/` and `/` is ignored

Can span multiple lines

Cannot be nested

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`/* This is a`

`multi-line comment`

spanning several lines */

let price = 100; / *Inline block comment* /